

SIMATS ENGINEERING ASSESSMENT TOOL 2

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COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.

5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.

4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1.Smart Campus Network (192.168.100.0/24)

School of Computing (/25)

Step 1: Subnet Mask = 255.255.255.128

Step 2: Block Size = 128

Network Address: 192.168.100.0

Host IP Range: 192.168.100.1 – 192.168.100.126

Broadcast Address: 192.168.100.127

Usable Hosts: $2^7 - 2 = 126$

School of Electronics (/26)

Step 1: Subnet Mask = 255.255.255.192

Step 2: Block Size = 64

Network Address: 192.168.100.128

Host IP Range: 192.168.100.129 – 192.168.100.190

Broadcast Address: 192.168.100.191

Usable Hosts: $2^6 - 2 = 62$

School of Mechanical Engineering (/27)

Step 1: Subnet Mask = 255.255.255.224

Step 2: Block Size = 32

Network Address: 192.168.100.192

Host IP Range: 192.168.100.193 – 192.168.100.222

Broadcast Address: 192.168.100.223

Usable Hosts: $2^5 - 2 = 30$

Remaining Unused IP Address Range

Unused Range: 192.168.100.224 – 192.168.100.255

2. Multinational Software Company (10.10.0.0/16) Software Development (/24)

Step 1: Subnet Mask = 255.255.255.0

Network Address: 10.10.0.0

Host IP Range: 10.10.0.1 – 10.10.0.254

Broadcast Address: 10.10.0.255

Usable Hosts: $2^8 - 2 = 254$

Quality Assurance (/25)

Step 1: Subnet Mask = 255.255.255.128

Network Address: 10.10.1.0

Host IP Range: 10.10.1.1 – 10.10.1.126

Broadcast Address: 10.10.1.127

Usable Hosts: 126

Human Resources (/26)

Step 1: Subnet Mask = 255.255.255.192

Network Address: 10.10.1.128

Host IP Range: 10.10.1.129 – 10.10.1.190

Broadcast Address: 10.10.1.191

Usable Hosts: 62

Executive Management (/27)

Step 1: Subnet Mask = 255.255.255.224

Network Address: 10.10.1.192

Host IP Range: 10.10.1.193 – 10.10.1.222

Broadcast Address: 10.10.1.223

Usable Hosts: 30

Remaining Unused IP Address Range

Unused Range: 10.10.1.224 – 10.10.255.255

3. IPv6 Address

Given Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

Step 1: Compress the IPv6 Address

Remove leading zeros and replace consecutive zero blocks with "::".

Compressed Address:

2001:db8:abcd::/64

Step 2: Expand the Address

Expanded Address:

2001:0db8:abcd:0000:0000:0000:0000:0000

Step 3: Network Prefix (/64)

The first four groups form the network prefix.

Network Prefix:

2001:db8:abcd:0::/64

Step 4: Total Host Addresses

Host bits = $128 - 64 = 64$

Total Hosts = 2^{64}

= 18,446,744,073,709,551,616

Step 5: Total Possible Host Addresses

Answer: 18,446,744,073,709,551,616 host addresses

5. Routing Table

Given Routing Table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP: 192.168.2.45

Step 1: Destination Subnet

The IP 192.168.2.45 belongs to

Subnet: 192.168.2.0/24

Step 2: Matching Routing Entry

192.168.2.0/24

Step 3: Next-Hop Network/Interface

Forward the packet through the router interface connected to the 192.168.2.0/24 network

Step 4: Default Route

If no route matches, use the

Default Route: 0.0.0.0/0

5. Administration LAN and Finance LAN Administration LAN (192.168.10.0/26)

Step 1: Network and Broadcast Address

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Step 2: Host IP Range

192.168.10.1 – 192.168.10.62

Step 3: Three Computer IP Addresses

Computer 1: 192.168.10.2

Computer 2: 192.168.10.3

Computer 3: 192.168.10.4

Step 4: Default Gateway

192.168.10.1

Finance LAN (192.168.10.64/26)

Step 1: Network and Broadcast Address

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Step 2: Host IP Range

192.168.10.65 – 192.168.10.126

Step 3: Three Computer IP Addresses

Computer 1: 192.168.10.66

Computer 2: 192.168.10.67

Computer 3: 192.168.10.68

Step 4: Default Gateway

192.168.10.65